

Leveraging Transformer Models for Extractive Question Answering in Criminal Report Analysis

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Abstract. Transformer-based models have significantly advanced Extractive Question Answering (EQA) in Natural Language Processing (NLP); yet their adaptation to legal document analysis remains understudied. This paper proposes a domain-adapted BERT model for Question Answering (QA) on Spanish-language robbery case reports from Ecuador’s Prosecutor’s Office (*Fiscalía General del Estado*), where the volume of reports and the complexity of legal language pose significant challenges for manual analysis. Our methodology employed Generative Artificial Intelligence (GAI) to synthesize a SQuAD-formatted question-answering dataset tailored to robbery case reports. We first evaluated the baseline performance of a pre-trained BERT model on this domain-specific dataset before conducting fine-tuning to adapt the model to the unique characteristics of legal documentation. The fine-tuned model demonstrated improved capability in processing criminal case information, achieving a final F1-score of 82.88, which represents a measurable enhancement over the baseline performance. These results demonstrate the effectiveness of fine-tuning transformer-based models for EQA tasks in legal language processing and underscore their potential to optimize information retrieval within the justice system. This could help to provide better service to the community in the near future by leveraging workloads.

Keywords: Legal Natural Language Processing · NLP · Natural Language Processing · transformers · LNLP · law.

1 Introduction

Transformer-based models have emerged as powerful tools in NLP, enabling more accurate and efficient extraction of relevant information from complex texts such as criminal reports. These models utilize self-attention mechanisms to weigh the importance of different words in a sentence, allowing for better context understanding and improved performance in tasks like QA. By fine-tuning these models on domain-specific datasets, researchers can enhance their ability to discern

pertinent details and provide precise answers to inquiries related to criminal incidents.

The implementation of EQA systems in legal contexts not only enhances the efficiency of information retrieval but also addresses challenges related to data variability and complexity inherent in criminal reports. As these models are fine-tuned on diverse datasets, they can improve their generalizability across different types of legal documents, ensuring that critical details are accurately captured regardless of nuances in language or structure [7]. Moreover, recent advancements in model architectures, such as the integration of dynamic masking techniques, have shown promise in further refining performance by enabling models to better adapt to varying levels of difficulty in questions posed about complex narratives [12]. This evolution highlights the potential for transformer-based models to facilitate more informed decision-making processes within the justice system, ultimately contributing to enhanced outcomes for legal practitioners and stakeholders alike.

The *Fiscalía General del Estado* (FGE), the Prosecutor’s Office in Ecuador, handles cases ranging from criminal investigations to public safety concerns, aiming to uphold justice and protect the rights of citizens. In this context, crimes of robbery are highly prevalent and present significant challenges for law enforcement agencies. Understanding the patterns and relationships behind these crimes can lead to more effective investigation processes. Additionally, it is important to analyze all the information gathered through crime reports. Many users also rely on this information to access insurance coverage for items lost during theft (e.g., cell phones, computers, automobiles, automobile parts, etc.). The volume of cases that need to be reviewed by prosecutors is substantial. Consequently, it is of interest to explore the use of Artificial Intelligence (AI) in designing and training models that can help reduce the high demand and improve public service.

Prior work has examined robbery classification using modern neural architectures, particularly through fine-tuned language models. Building on this foundation, our study explores the potential of EQA systems for processing criminal reports. This methodology could significantly aid legal authorities by extracting actionable insights for investigations, while also enabling comprehensive crime statistics analysis. To the best of our knowledge, this work presents a novel application of transformer-based models employing a curated Spanish-language SQuAD dataset from the Ecuadorian legal domain.

This paper is organized as follows: Section 2 surveys existing applications of Artificial Intelligence (AI) in legal domains, with a focus on question answering (QA) tasks. Section 3 describes our methodology, including dataset construction and model training. Section 4 presents the evaluation of our fine-tuned model. Finally, Section 5 summarizes our main findings and outlines opportunities for future work.

2 Literature Review

Large Language Models (LLMs) represent a category of Deep Learning (DL) models that integrate transformer architectures with large-scale pre-training on extensive text corpora. Trained in a self-supervised fashion, these models capture detailed linguistic patterns and can subsequently be fine-tuned to address domain-specific challenges in Legal NLP (LNLP). For instance, one influential model is described in [3], and another is presented in [14], where OpenAI reported that the model achieved performance at the 90th percentile on the Uniform Bar Exam, a standardized examination used to evaluate the competency of prospective attorneys. However, subsequent analysis by [10] indicates that GPT-4’s actual performance ranges from the 48th to the 62nd percentile. These authors critique the imprecision and lack of transparency in OpenAI’s report, arguing that such shortcomings may hinder the development of safe AI and lead to misconceptions regarding the model’s true capabilities in handling complex legal tasks.

Legal Question Answering (LQA) is a complex task that entails addressing queries related to legal issues, necessitating not only the analysis of extensive legal corpora but also their nuanced interpretation [2]. A wide range of models and algorithms have been explored for QA in NLP, such as *Recurrent Neural Networks (RNNs)*, *Long Short-Term Memory (LSTM)*, and *Convolutional Neural Networks (CNN)*. Additionally, information retrieval approaches, including *Term Frequency-Inverse Document Frequency (TF-IDF)*, are frequently utilized. Both extractive and generative methodologies have demonstrated effectiveness in addressing NLP-based QA tasks [9].

It is important to clarify the primary distinction between these two approaches. Extractive QA identifies and retrieves a span of text directly from the given context; given a tuple composed of the tokenized question q and the context c , the model predicts the start and end positions of the answer. In contrast, generative QA synthesizes responses using encoder-decoder architectures, generating answers in an autoregressive manner [9]. According to a systematic evaluation of QA models by [9], generative readers generally outperform extractive readers with longer contexts, providing more fluent and comprehensive answers, whereas extractive readers are more effective when the context is limited and exhibit better out-of-domain generalization.

2.1 Large Language Models and Generative AI

GAI refers to a class of AI models that learn the underlying data distribution $p_{data}(\mathbf{x})$ to generate novel samples (e.g., text, images, or audio) through an estimated model distribution $p_{model}(\mathbf{x})$ [4]. Modern AI models are considered Foundational Models, which represent a class of general-purpose models capable of handling diverse data modalities and supporting multiple AI tasks [6]. Notable examples include the Llama 3 family of models, which are foundational language models featuring multilingual capabilities, programming proficiency, reasoning skills, and tool integration [6]. Equation (1) represents the goal of GAI, which is

to learn a model distribution that accurately captures the underlying patterns in the unknown probabilistic distribution of observations (i.e., data), thereby enabling the generation of new observations by sampling the trained model.

$$p_{model}(\mathbf{x}) \rightarrow p_{data}(\mathbf{x}) \quad (1)$$

LLMs are a specialized subset of foundational AI and GAI models, with a primary focus on text-based tasks. These models leverage transformer architectures [11,6] and undergo extensive pre-training on textual corpora. Notably, LLMs exhibit exceptional capabilities in natural language understanding and complex problem-solving (i.e., General Purpose Task Solver) [17], rendering them particularly effective for sophisticated linguistic tasks. The basic working of an LLMs can be thought of as a next-word prediction problem, where given a sequence of tokens $\mathbf{x}_{1:t} = (x_1, \dots, x_t)$, an LLMs defines a conditional probability distribution over the vocabulary $\mathcal{V}$:

$$P(x_{t+1}|\mathbf{x}_{1:t};\theta) = \text{softmax}(\mathbf{W} \cdot \mathbf{h}_t + \mathbf{b}) \quad (2)$$

θ denotes the model parameters, and $\mathbf{h}_t$ is obtained by several layers of transformer blocks that process an embedded representation of the tokenized text:

$$\mathbf{h}_t = \text{Transformer}(\mathbf{x}_{1:t};\theta) \quad (3)$$

2.2 Related Works

There is significant interest within both the scientific community and legal institutions in studying the applications of AI in this expansive field. The current state of the art in NLP with DL primarily focuses on the use of transformer-based architectures, which have demonstrated superior performance across various NLP tasks.

In [16], the deployment of a RoBERTa Base QA classification model specifically for legal questions is discussed. It compares the performance of this transformer model against a traditional linear SVM in the legal domain, using the PRIVACYQA dataset. The results indicate that RoBERTa significantly outperforms the SVM, achieving a 31% improvement in F1-score and a 41% improvement in Mean Reciprocal Rank, demonstrating the effectiveness of transformers in enhancing answer retrieval for legal inquiries. In [15], the researchers focus on a Transformer-based neural network for answer selection in general QA systems, not specifically in the legal domain. While Transformers are effective in extracting global information and improving answer selection, the research does not address their application in law-related QA. For insights on using Transformers in legal contexts, further research specific to LNL applications would be necessary. For information retrieval, [8] discuss the use of transformer-based approaches for legal information retrieval and entailment, specifically in the context of the COLIEE competition. For case law retrieval (Task 1), a sentence-transformer model was employed to create numeric representations of case paragraphs, facilitating

similarity assessments. In entailment (Task 4), a fine-tuned DeBERTa large language model was utilized, achieving third place among eight teams. These methods demonstrate the effectiveness of transformers in addressing LQA challenges. Another area of interest is predicting legal judgments. In [5] six transformer-based models (BERT, XLNet, RoBERTa, DeBERTa, ELECTRA, and BigBird) are evaluated in predicting legal judgments using the ILDC dataset. It highlights the potential of these models in automating legal decision-making, with newer models achieving up to 80% accuracy compared to 76% for older models. This comparative analysis underscores the advancements in NLP technology and its applicability in the legal domain, suggesting a promising future for QA in law through transformer models. Finally, in [13] the use of transformers to generate legal text is addressed, indicating their potential to assist in legal practice. The findings suggest that these models can effectively generate judicial opinions and help differentiate between human and machine-generated legal texts, which could be beneficial for tasks like drafting contracts and briefs, although direct application to QA is not discussed.

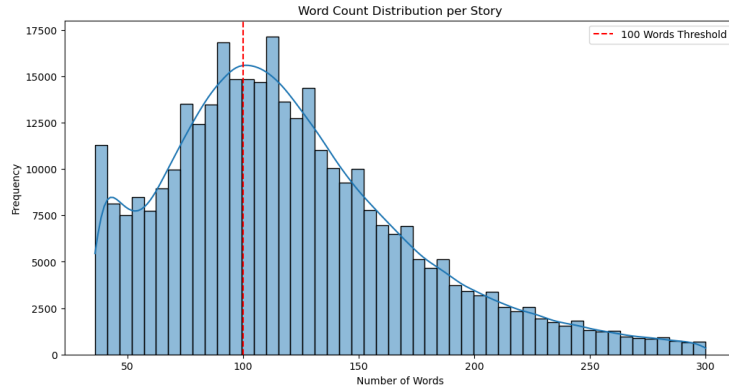
In this section, we briefly discuss foundational concepts and present research related to our work. It is noteworthy that this study introduces a novel application of transformer-based models. To the best of our knowledge, there is currently no evidence of similar studies that simultaneously address both Spanish and legal texts.

3 Methodology

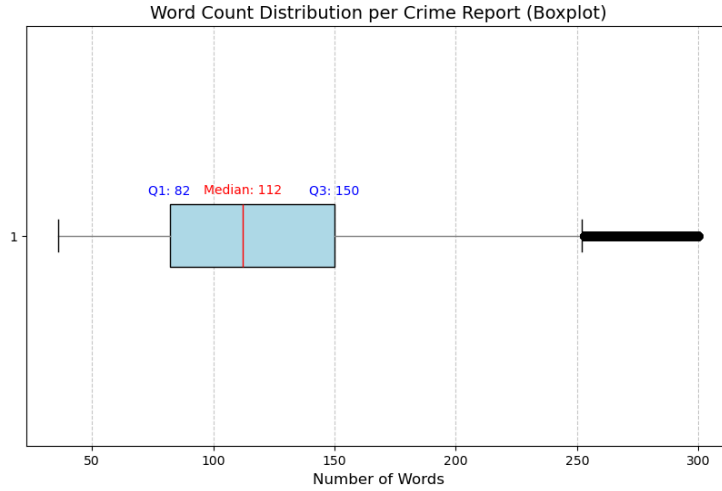
In this paper, we present a fine-tuned BERT-based model for EQA on Spanish-language criminal reports. The model, adapted from a pre-trained Spanish BERT architecture optimized for LQA tasks, was developed to support information extraction from robbery crime cases. Our methodology consists of three phases: (1) dataset generation, (2) pre-trained baseline model performance, and (3) model fine-tuning.

3.1 Robbery Crime Question Answering Dataset

Data Collection We constructed our dataset using a set of unlabeled robbery crime reports. To create a QA dataset in the SQuAD format, we employed GAI. Statistical analysis of the collected reports revealed an average length (mean) of 112 words, and an upper bound of 150 words. Consequently, we included reports x_i satisfying $82 \leq \Gamma(x_i) \leq 150$, where $\Gamma : x_i \rightarrow \mathbb{N}$ is a function that returns the word count of report x_i . This filtering criterion effectively removed statistical outliers while maintaining the core distribution. Figure 1 presents the distribution of word counts in the original dataset through both a histogram and box plot visualizations, representing the complete set of 174,594 samples.



(a) Original Dataset Histogram



(b) Box plot of Words Length in Original Dataset

Fig. 1: Distribution of word counts in collected data visualized through (a) a histogram and (b) a box plot

Generative Artificial Intelligence Prompt Engineering For this research, we used Meta-LLAMA-3-8B-Instruct[1], an instruction-tuned variant of Meta’s LLAMA 3 8B parameter language model, optimized for conversational and task-oriented interactions with multilingual capabilities. Built on a decoder-only TRANSFORMER architecture, it leverages supervised fine-tuning (SFT) and reinforcement learning from human feedback (RLHF) to align responses with user intent. The model is pre-trained on a diverse corpus of publicly available text data, with enhancements in reasoning, coding, and safety mitigation compared to its predecessors.

Prompt engineering is the process of crafting a suitable prompt to drive the LLMs to solve a complex task, using natural language text [17]. In this work, the LLM functions as an expert data science assistant with the goal of constructing a dataset in the SQuAD format in Spanish. For this purpose, the prompt receives the content of the crime report as the `context` and each of the questions from Table 1 as the `question`. The LLM was also instructed to retrieve a `json` format output with the variables: `context`, `question`, `answer_text`, `answer_start`, `answer_end`, and `impossible_find_answer`. The latter was calculated by finding the `answer_text` in the context. In the case that no answer was possible to be retrieved, either because there is no answer or because the LLM failed, the variable `impossible_find_answer` was set to `True`.

Table 1: Proposed Questions for the Dataset

Question
What objects were stolen?
On what date did the incident occur?
At what time did the robbery happen?
At what address or between which streets did the robbery/incident occur?
What was the dollar value of the stolen or taken objects?

Hugging Face’s Inference API processed samples at a speed of 6.4 samples per minute. A total of 1,544 samples were obtained after four hours. Our evaluation revealed two key limitations of the LLM: (1) incomplete retrieval of answers for certain queries, and (2) sporadic generation of irrelevant content. These results suggest that manual verification of the generated dataset can aid in improving quality. However, the engineered prompt allowed for the identification of problems in generated samples.

Dataset Quality After collecting the initial 1,544 samples, which generated 7,705 question-answer pairs at five questions per context, we conducted quality assessment using the `impossible_find_answer` flag. This process identified approximately 3,000 invalid cases, leading us to retain only samples where `impossible_find_answer` is `False`. Through histogram and box plot analysis of answer lengths, we established an upper limit of 17 words for valid responses. The final curated dataset consisted of 4,572 samples, all containing at least one word and meeting our validity criteria, suitable for model training and evaluation.

Dataset Train and Test Split The dataset was partitioned into training and testing sets, with 20% of the samples allocated for evaluation. The remaining 80% were used for training. This approach is a fundamental machine learning technique that helps to evaluate the model’s performance and generalization capability. Table 2 summarizes the sample sizes for each subset. The split was performed randomly.

Table 2: Dataset Train and Test Split

Partition	# Samples
Train	3,657
Test	915
Total	4,572

3.2 Model Fine-Tuning

The `mrm8488/bert-base-spanish-wwm-cased-finetuned-spa-squad2-es` model was selected for fine-tuning because it is a model trained for Spanish and QA tasks. As a matter of fact, it was tested without training and it performed with an F1-Score of 81.70.

To fine-tune the model, both the question and the answer need to be tokenized. As a result of this process, the indices indicating the start and end of the answer must be recalculated based on the tokenized embedding. In the case of long text, truncation should be applied only to the context, and a window of 128 tokens was used to avoid losing the answer during truncation. However, in this scenario, since most of the current samples have no more than 100 words in the context, this issue is unlikely to occur; nonetheless, it has been addressed in the program.

In order to train the model, the parameters shown in Table 3 were used. A small learning rate is recommended for fine-tuning in order to reuse the pre-training from the original model and adapt the new model to the target dataset. Our results prove that it is possible to achieve a better performance on the target domain with the fine-tuned model rather than using the pre-trained model. Although the difference in performance is small, it could be related to the fact that we were not able to use the whole dataset.

Table 3: Training Parameters

Parameter	Value
Strategy	epoch
Learning rate	2×10^{-5}
Batch Size	64
Epochs	10
Weight Decay	0.01
FP16	True
Metric for best model F1 (eval)	

4 Results

In this section, we present the results of fine-tuning the model, along with the performance of the pre-trained model on the test set. This comparison was conducted to determine whether the fine-tuned model demonstrated any improvements over the original pre-trained model. Table 4 shows the performance of

the pre-trained model, while Table 5 shows the performance of the fine-tuned model. It is worth noting that the latter improves upon the former. The fine-tuned model achieved an F1 score of 82.88 on the evaluation dataset. In this case, we use `squad_v2` which contains both the F1-SCORE and EXACT MATCH. A function was prepared in order to be used by the Trainer Class.

Table 4: Pre-trained Model Performance

Metric	Value
eval_loss	0.9102
eval_model_preparation_time	0.0077
eval_exact	53.5519
eval_f1	81.7041
eval_total	915.0000
eval_HasAns_exact	53.5519
eval_HasAns_f1	81.7041
eval_HasAns_total	915.0000
eval_best_exact	53.5519
eval_best_exact_thresh	0.0000
eval_best_f1	81.7041
eval_best_f1_thresh	0.0000
eval_runtime	12.3982
eval_samples_per_second	73.8010
eval_steps_per_second	1.2100

Table 5: Fine-Tuned Model Performance

Metric	Value
eval_loss	1.0088
eval_model_preparation_time	0.0077
eval_exact	55.7377
eval_f1	82.8805
eval_total	915.0000
eval_HasAns_exact	55.7377
eval_HasAns_f1	82.8805
eval_HasAns_total	915.0000
eval_best_exact	55.7377
eval_best_exact_thresh	0.0000
eval_best_f1	82.8805
eval_best_f1_thresh	0.0000
eval_runtime	12.8109
eval_samples_per_second	71.4240
eval_steps_per_second	1.1710
epoch	10.0000

5 Conclusions and Future Works

In this study, we demonstrated the effectiveness of transformer-based models for EQA within the context of criminal report analysis. Our fine-tuned model

showed a notable improvement, achieving an F1 score of 82.88 compared to the pre-trained model’s score of 81.70. This indicates the potential for enhanced performance through domain-specific fine-tuning.

The application of our model to robbery crime reports could facilitate the extraction of crucial information, supporting legal practitioners in their investigations and reporting. However, the study also highlighted some limitations, including the model’s occasional inability to retrieve accurate answers and the need for manual verification of results.

Future work will aim to address these limitations by exploring additional advanced techniques and incorporating manual labeling of the dataset by expert peers. Additional computational resources will also be required to process larger datasets. Furthermore, the integration of AI into the legal domain holds significant promise for more efficient case management and resource allocation, ultimately contributing to a safer community. It is also of interest to investigate other NLP techniques within the Ecuadorian legal context, such as Legal Named Entity Recognition (NER) and Legal Document Summarization (LDS). We encourage further research and collaboration to refine these approaches and harness the full potential of AI in supporting the justice system.

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